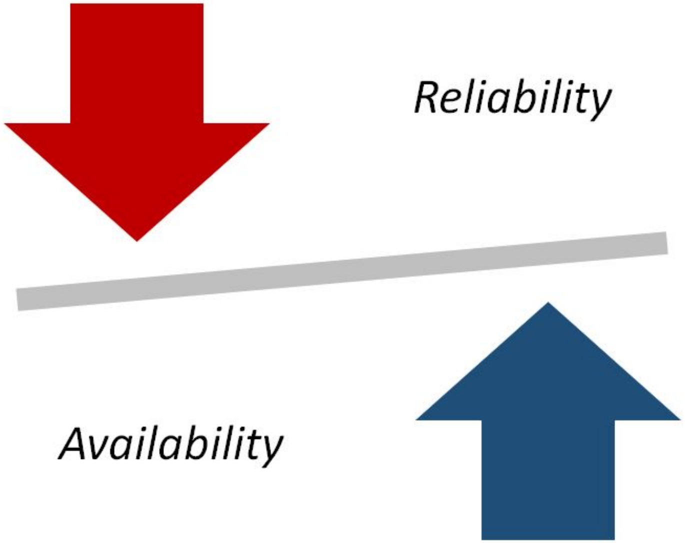
Reliability & Fault Tolerance

System Design Crash Course: Quick Revision Before Interviews

Rahul Rajat Singh

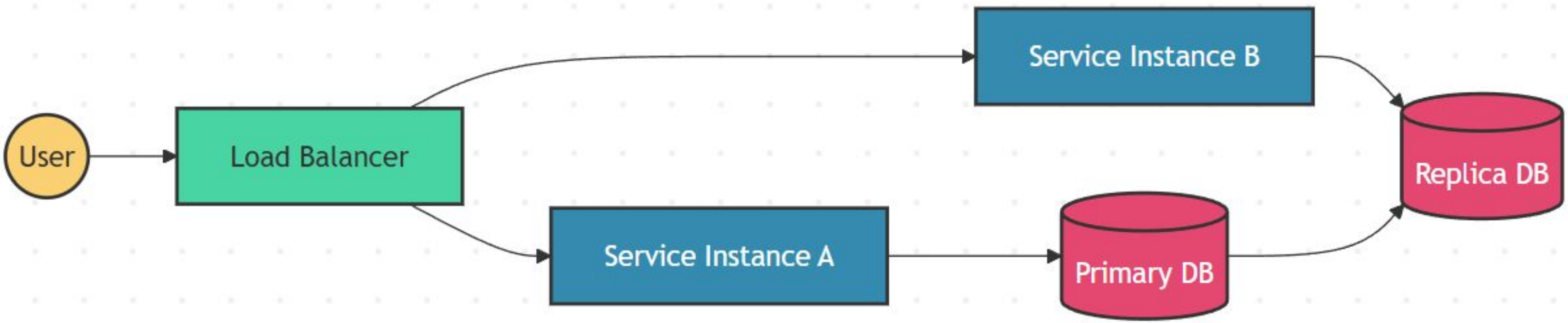
Reliability vs Availability

- Reliability: correctness over time
- Availability: uptime and access
- Failures vs outages
- Retry hides failure
- Consistency trade-offs



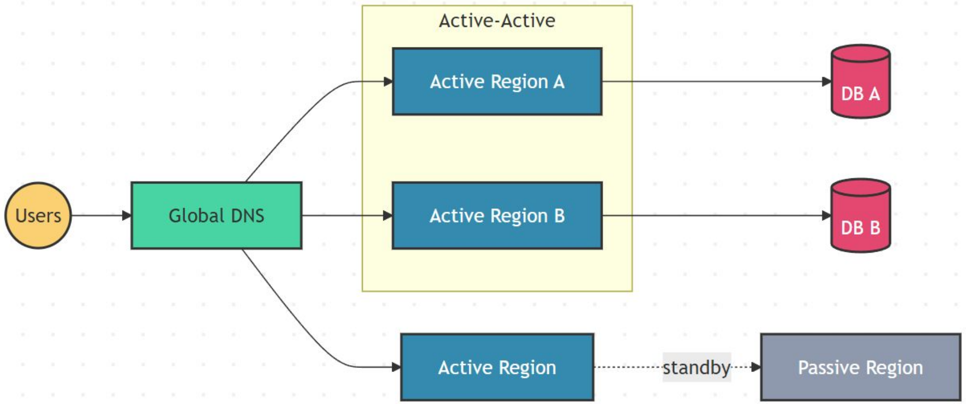
High Availability

- Eliminate single points of failure
- Redundant components everywhere
- Automatic failover required
- Health checks and load balancing
- Availability over consistency sometimes



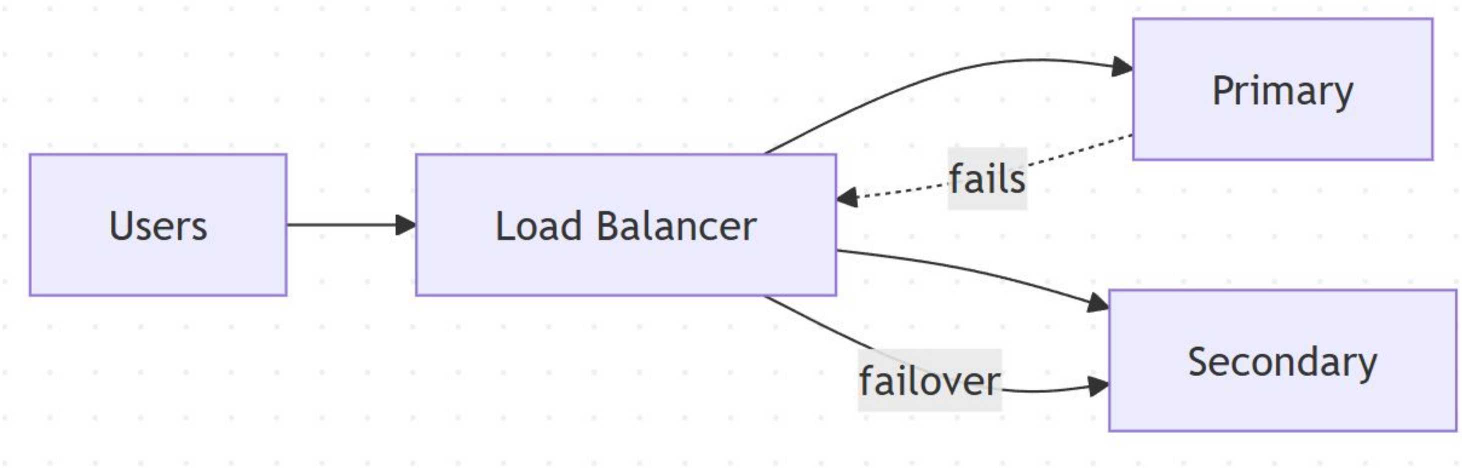
Active-Active vs Active-Passive

- Active-Active: serve traffic everywhere
- Active-Passive: standby takes over
- Failover speed differs
- Cost vs utilization trade-off
- Consistency complexity varies



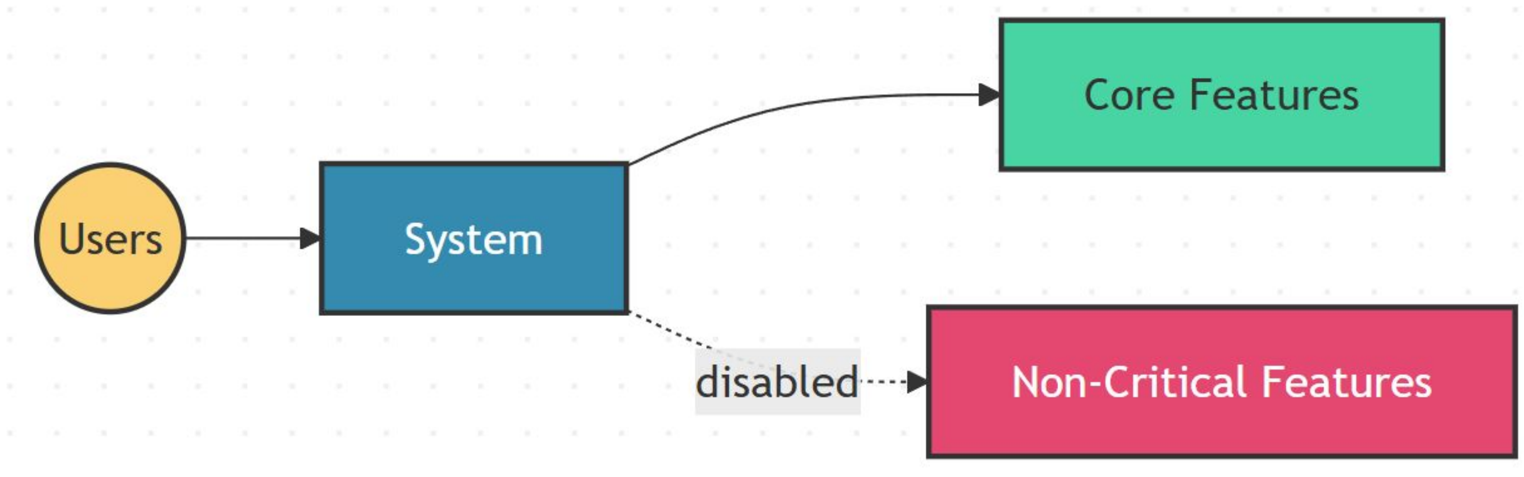
Failover Mechanisms

- Detect failure fast
- Decide primary vs secondary
- Redirect traffic automatically
- Minimize recovery time
- False positives hurt availability



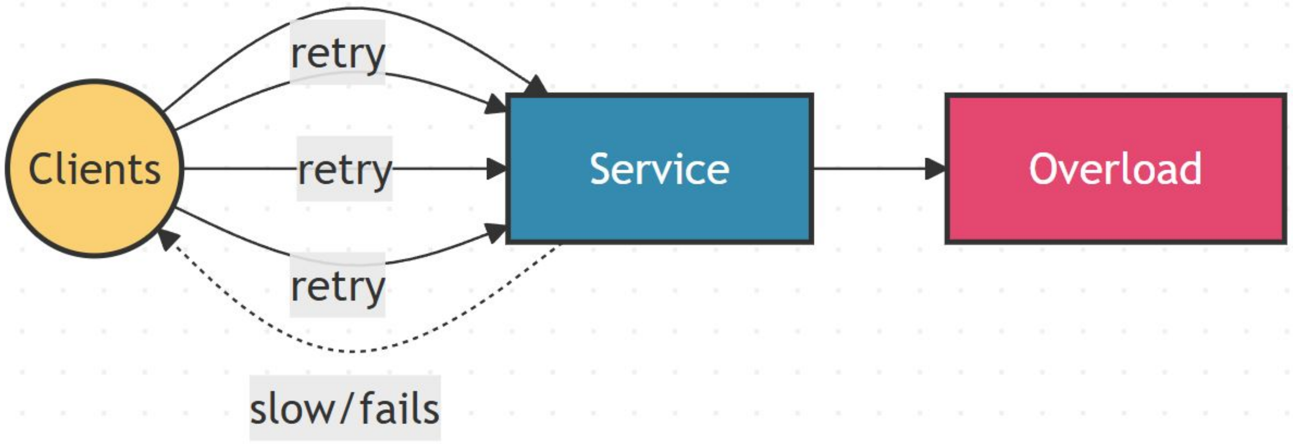
Graceful Degradation

- Serve core features only
- Shed non-critical functionality
- Partial failure, not total
- Protect critical paths
- User experience prioritized



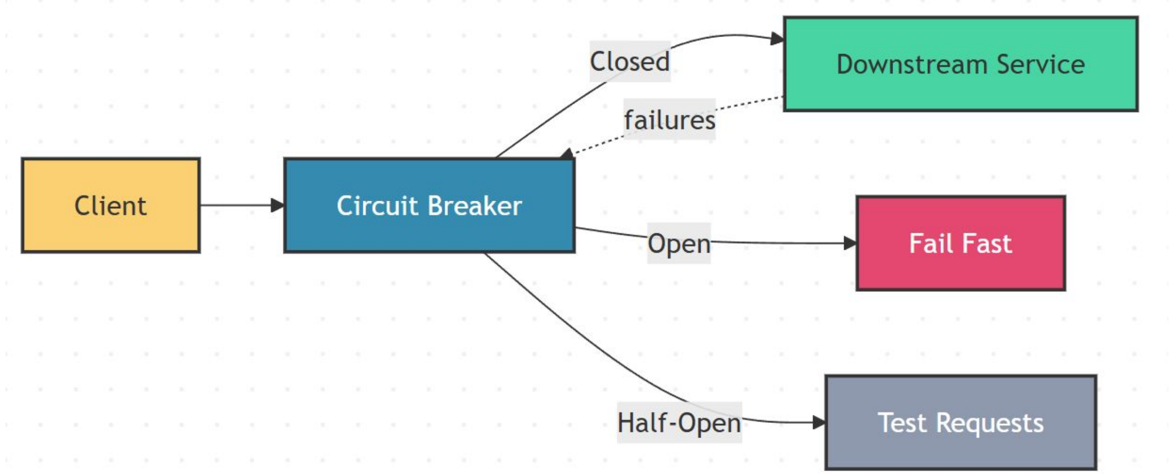
Retries and Retry Storms

- Retries mask transient failures
- Uncontrolled retries amplify load
- Retry storms cause cascading failures
- Backoff and jitter required
- Retries are not always safe



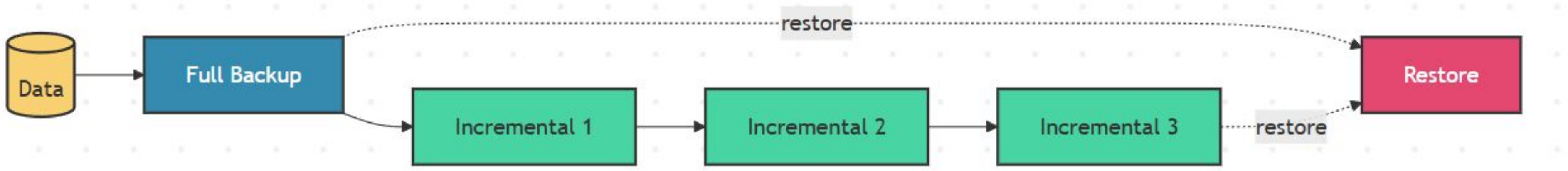
Circuit Breakers

- Monitor failures continuously
- Closed, open, half-open states
- Fail fast when unhealthy
- Protect downstream services
- Availability over immediate retries



Backup Strategies

- Protect against data loss
- Full vs incremental backups
- Recovery time matters
- Storage and cost trade-off
- Test restores regularly



RTO vs RPO

- RTO: recovery time objective
- RPO: acceptable data loss
- Time vs data trade-off
- Drives backup strategy
- Business-defined metrics